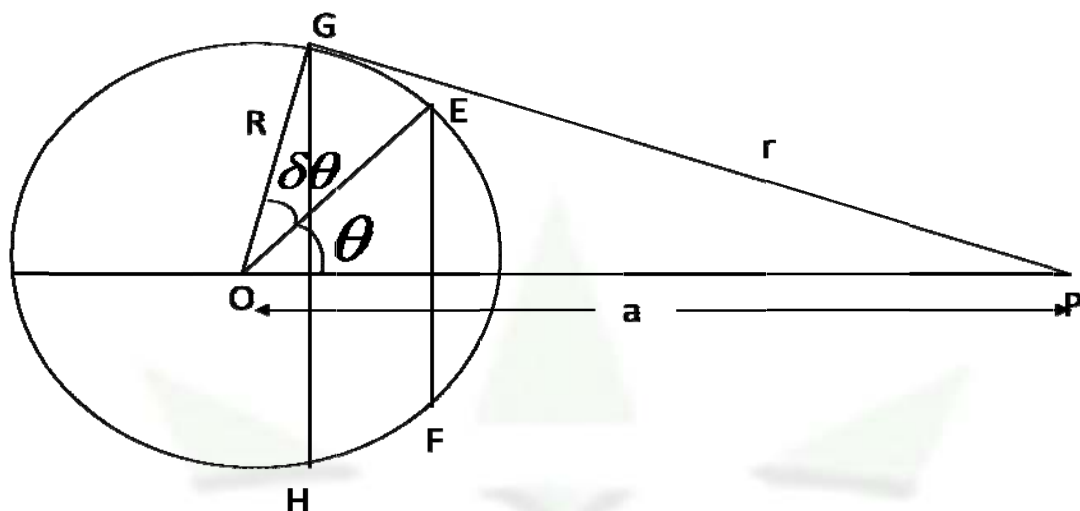




Gravitation – Gravitational Potential Due To A Hollow Shell

Gravitational potential at a point due to a thin hollow shell (spherical):



In the figure the circle represents the hollow shell P is the given point.

Given:

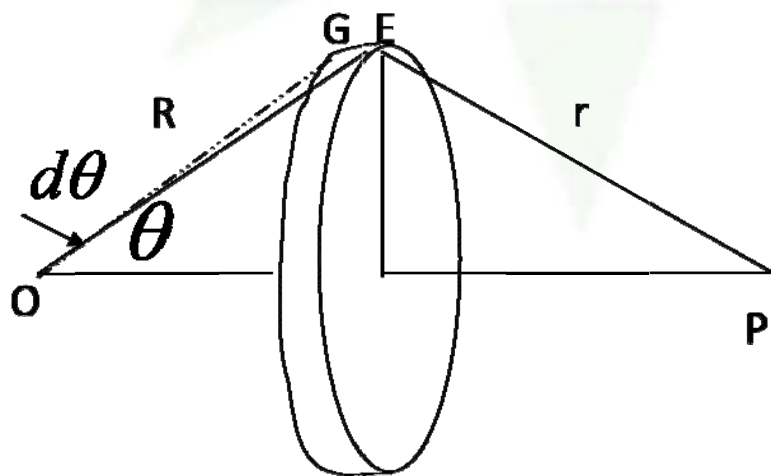
M & R = mass and radius of the hollow shell

$a = OP$ = distance of the given point P from the center of the shell.

Surface area of the shell $= 4\pi R^2$

Mass per unit area of the shell $= \frac{M}{4\pi R^2}$

Join OP let us cut the shell by two imaginary planes EF & GH perpendicular to the line OP and separated by a vanishingly small distance. The cut section appears like an elementary ring & P is a point on the axis of the ring.

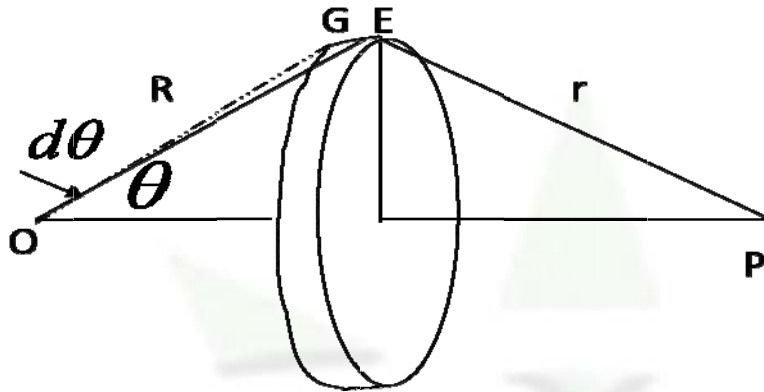




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Join EP let $EP = r$, distance of the point P will be same equal to r from every point on the circumference of the ring.

$$\angle POE = \theta, \angle EOG = d\theta$$



From the figure the radius of the elementary ring $= R \sin \theta$

The thickness of the elementary ring EG $= R d\theta$

The surface area of this elementary ring = circumference x thickness

$$= 2\pi R \sin \theta \times R d\theta$$

$$= 2\pi R^2 \sin \theta d\theta$$

$$\begin{aligned} \text{Mass of the elementary ring} &= \frac{M}{4\pi R^2} \times 2\pi R^2 \sin \theta d\theta \\ &= \frac{M}{2} \sin \theta d\theta \end{aligned}$$

Since the given point P is equidistance from every point on the circumference of the ring potential at P due to elementary ring

$$dv = - \frac{G \times (\text{mass})}{r}$$

$$dv = - \frac{GM \sin \theta d\theta}{2r} \longrightarrow (1)$$

Since the given shell can be assumed to be made up of many such elementary rings. Potential at P due to the whole shell can be obtained by integrating equation (1), but since equation (1) contains two variables r and θ hence it is to be represented in terms of one variable.

$$\text{From } \triangle EOP \text{ using properties of triangle } r^2 = R^2 + a^2 - 2Ra \cos \theta \longrightarrow (2)$$

$$\text{Differentiating equation (2)} \quad 2rdr = 0 + 0 - 2Ra(-\sin \theta d\theta)$$

$$\frac{\sin \theta d\theta}{r} = \frac{dr}{Ra} \longrightarrow (3)$$

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Putting equation (3) in (1)

$$dv = -\frac{GM}{2} \frac{dr}{Ra} \longrightarrow (4)$$

Case I: The given point P lies outside the shell ($a > R$), in this case from the figure we find that the limits of integration $r = (a - R)$ to $r = (a + R)$

$$\begin{aligned} \text{Potential at P} \quad V &= \int dv = \int_{a-R}^{a+R} -\frac{GM}{2Ra} dr \\ V &= -\frac{GM}{2Ra} [r]_{a-R}^{a+R} = -\frac{GM}{2Ra} [a+R - a+R] \\ V &= -\frac{GM}{a} \longrightarrow (5) \end{aligned}$$

If we imagine a point mass M (equal to the mass of the shell) at O the center of the shell then potential at P.

$$V = -\frac{GM}{a} \longrightarrow (6)$$

Comparing equation (5) and (6) we can say that to find Gravitational potential at an external point due to a thin hollow shell we can assume the mass of the shell to be concentrated at its center.

Case II: The given point P lies on the surface of the shell ($a = R$) from the figure in this case the limits of integration is $r=0$ to $r = 2R$

Potential at P:

$$\begin{aligned} V &= \int dv = \int_0^{2R} -\frac{GM}{2R.R} dr \\ V &= -\frac{GM}{R} \longrightarrow (7) \end{aligned}$$

The gravitational potential at any point on the circumference of a shell is constant.

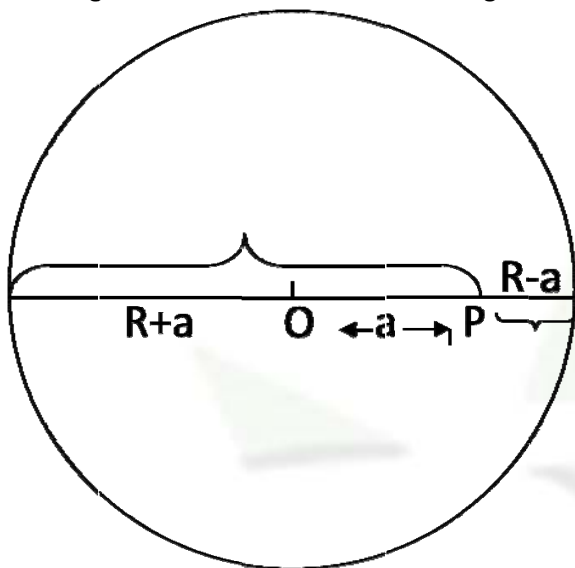
Hence Intensity ($I = \frac{dv}{dr}$) at every point on the surface of a shell is zero.



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Case III: The given point P lies inside the shell $a < R$

From figure in this case the limits of integration $R = R - a$ to $r = R + a$



$$V = \int_{R-a}^{R+a} dv = \int_{R-a}^{R+a} -\frac{GM}{2Ra} dr = -\frac{GM}{2Ra} [r]_{R-a}^{R+a}$$

$$V = -\frac{GM}{R} = \text{Constant}$$

Thus the gravitational potential at any point inside shell is constant & is same as the potential on the surface of the shell. Since the potential is constant the gravitational intensity at any point inside the shell is zero.